

TikTok Video View Summary

Setup & Data Import

This section loads the necessary R libraries and imports the raw TikTok video view dataset for our analysis.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v forcats   1.0.1      v readr      2.2.0
```

```
v ggplot2   4.0.3      v stringr    1.6.0
```

```
v lubridate 1.9.5      v tibble     3.3.1
```

```
v purrr     1.2.2      v tidyr      1.3.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
views <- read.csv("data/raw/video_view.csv")
```

Data Inspection

Here we check the size and structure of our data to make sure everything looks right.

```
count_rows <- nrow(views)
print(count_rows)
```

```
[1] 15000
```

```
str(views)
```

```
'data.frame':  15000 obs. of  16 variables:
 $ video_id          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ creator_id        : int  66 62 176 22 39 148 158 30 200 19 ...
 $ video_length_sec   : int  14 47 37 29 28 36 30 34 25 48 ...
 $ published_at       : int  1767297237 1768475971 1766815457 1755689136 17626964...
 $ video_url          : chr  "https://www.tiktok.com/@olivialive2824/video/9000000...
 $ impressions_n      : int  4 0 14 67 25 14 23 19 6 86 ...
 $ watched_n          : int  3 0 10 49 18 7 19 14 4 64 ...
 $ watch_rate         : num  0.75 NA 0.714 0.731 0.72 ...
 $ avg_watch_seconds_when_watched: num  10.7 NA 18.6 15.4 14.4 ...
 $ total_watch_seconds : int  32 0 186 754 260 133 322 273 73 1517 ...
 $ avg_watch_share     : num  0.571 NA 0.359 0.388 0.371 ...
 $ distinct_watchers_n : int  3 0 10 43 15 7 19 14 4 59 ...
 $ first_impression_at : int  1768287244 NA 1766868873 1755661745 1762792600 17650...
 $ last_impression_at  : int  1769439546 NA 1769741156 1769429852 1769860569 17691...
 $ first_watch_at      : int  1768287244 NA 1766868873 1755661745 1762792600 17650...
 $ last_watch_at       : int  1769439546 NA 1769741156 1769429852 1769860569 17691...
```

```
head(views)
```

	video_id	creator_id	video_length_sec	published_at
1	1	66	14	1767297237
2	2	62	47	1768475971
3	3	176	37	1766815457
4	4	22	29	1755689136

5	5	39	28	1762696418
6	6	148	36	1764970157

	video_url
1	https://www.tiktok.com/@olivialive2824/video/9000000000000000001
2	https://www.tiktok.com/@isabellabrown/video/9000000000000000002
3	https://www.tiktok.com/@isabellawalker/video/9000000000000000003
4	https://www.tiktok.com/@lucastaylor/video/9000000000000000004
5	https://www.tiktok.com/@avabrown/video/9000000000000000005
6	https://www.tiktok.com/@harperwalker_1/video/9000000000000000006

	impressions_n	watched_n	watch_rate	avg_watch_seconds_when_watched
1	4	3	0.7500000	10.66667
2	0	0	NA	NA
3	14	10	0.7142857	18.60000
4	67	49	0.7313433	15.38776
5	25	18	0.7200000	14.44444
6	14	7	0.5000000	19.00000

	total_watch_seconds	avg_watch_share	distinct_watchers_n	first_impression_at
1	32	0.5714286	3	1768287244
2	0	NA	0	NA
3	186	0.3590734	10	1766868873
4	754	0.3880597	43	1755661745
5	260	0.3714286	15	1762792600
6	133	0.2638889	7	1765069292

	last_impression_at	first_watch_at	last_watch_at
1	1769439546	1768287244	1769439546
2	NA	NA	NA
3	1769741156	1766868873	1769741156
4	1769429852	1755661745	1769429852
5	1769860569	1762792600	1769860569
6	1769101217	1765069292	1769101217

```
summary(views)
```

video_id	creator_id	video_length_sec	published_at
Min. : 1	Min. : 1.00	Min. :10.00	Min. :1.754e+09
1st Qu.: 3751	1st Qu.: 48.00	1st Qu.:23.00	1st Qu.:1.758e+09
Median : 7500	Median : 97.00	Median :30.00	Median :1.762e+09
Mean : 7500	Mean : 98.06	Mean :30.08	Mean :1.762e+09
3rd Qu.:11250	3rd Qu.:146.00	3rd Qu.:37.00	3rd Qu.:1.766e+09
Max. :15000	Max. :200.00	Max. :60.00	Max. :1.770e+09

video_url	impressions_n	watched_n	watch_rate
-----------	---------------	-----------	------------

Length	:15000	Min.	: 0.00	Min.	: 0.00	Min.	:0.0000
N.unique	:15000	1st Qu.:	9.00	1st Qu.:	6.00	1st Qu.:	0.6250
N.blank	: 0	Median	: 22.00	Median	: 15.00	Median	:0.6957
Min.nchar:	58	Mean	: 29.76	Mean	: 20.45	Mean	:0.6947
Max.nchar:	70	3rd Qu.:	41.00	3rd Qu.:	28.00	3rd Qu.:	0.7679
		Max.	:230.00	Max.	:158.00	Max.	:1.0000
						NAs	:384
avg_watch_seconds_when_watched		total_watch_seconds		avg_watch_share			
Min.	: 1.00	Min.	: 0.0	Min.	:0.0000		
1st Qu.:	11.75	1st Qu.:	92.0	1st Qu.:	0.3148		
Median	:16.28	Median	: 226.0	Median	:0.3747		
Mean	:16.42	Mean	: 336.5	Mean	:0.3794		
3rd Qu.:	20.56	3rd Qu.:	464.0	3rd Qu.:	0.4352		
Max.	:52.00	Max.	:3945.0	Max.	:1.0000		
NAs	:536			NAs	:384		
distinct_watchers_n		first_impression_at		last_impression_at			
Min.	: 0.00	Min.	:1.754e+09	Min.	:1.754e+09		
1st Qu.:	6.00	1st Qu.:	1.758e+09	1st Qu.:	1.769e+09		
Median	: 15.00	Median	:1.762e+09	Median	:1.770e+09		
Mean	: 19.35	Mean	:1.762e+09	Mean	:1.769e+09		
3rd Qu.:	27.00	3rd Qu.:	1.766e+09	3rd Qu.:	1.770e+09		
Max.	:135.00	Max.	:1.770e+09	Max.	:1.770e+09		
		NAs	:384	NAs	:384		
first_watch_at		last_watch_at					
Min.	:1.754e+09	Min.	:1.754e+09				
1st Qu.:	1.758e+09	1st Qu.:	1.769e+09				
Median	:1.762e+09	Median	:1.770e+09				
Mean	:1.762e+09	Mean	:1.769e+09				
3rd Qu.:	1.766e+09	3rd Qu.:	1.770e+09				
Max.	:1.770e+09	Max.	:1.770e+09				
NAs	:384	NAs	:384				

```
colSums(is.na(views))
```

video_id	creator_id
0	0
video_length_sec	published_at
0	0
video_url	impressions_n
0	0
watched_n	watch_rate
0	384

avg_watch_seconds_when_watched	total_watch_seconds
536	0
avg_watch_share	distinct_watchers_n
384	0
first_impression_at	last_impression_at
384	384
first_watch_at	last_watch_at
384	384

```
names(views)
```

[1] "video_id"	"creator_id"
[3] "video_length_sec"	"published_at"
[5] "video_url"	"impressions_n"
[7] "watched_n"	"watch_rate"
[9] "avg_watch_seconds_when_watched"	"total_watch_seconds"
[11] "avg_watch_share"	"distinct_watchers_n"
[13] "first_impression_at"	"last_impression_at"
[15] "first_watch_at"	"last_watch_at"

```
glimpse(views)
```

Rows: 15,000

Columns: 16

\$ video_id	<int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ~
\$ creator_id	<int> 66, 62, 176, 22, 39, 148, 158, 30, 200, ~
\$ video_length_sec	<int> 14, 47, 37, 29, 28, 36, 30, 34, 25, 48, ~
\$ published_at	<int> 1767297237, 1768475971, 1766815457, 175~
\$ video_url	<chr> "https://www.tiktok.com/@olivialive2824~
\$ impressions_n	<int> 4, 0, 14, 67, 25, 14, 23, 19, 6, 86, 4, ~
\$ watched_n	<int> 3, 0, 10, 49, 18, 7, 19, 14, 4, 64, 3, ~
\$ watch_rate	<dbl> 0.7500000, NA, 0.7142857, 0.7313433, 0.~
\$ avg_watch_seconds_when_watched	<dbl> 10.66667, NA, 18.60000, 15.38776, 14.44~
\$ total_watch_seconds	<int> 32, 0, 186, 754, 260, 133, 322, 273, 73~
\$ avg_watch_share	<dbl> 0.5714286, NA, 0.3590734, 0.3880597, 0.~
\$ distinct_watchers_n	<int> 3, 0, 10, 43, 15, 7, 19, 14, 4, 59, 3, ~
\$ first_impression_at	<int> 1768287244, NA, 1766868873, 1755661745, ~
\$ last_impression_at	<int> 1769439546, NA, 1769741156, 1769429852, ~
\$ first_watch_at	<int> 1768287244, NA, 1766868873, 1755661745, ~
\$ last_watch_at	<int> 1769439546, NA, 1769741156, 1769429852, ~

General Summary

This section gives an overview of the main numbers, such as average video length, impressions, and watch rates.

```
overview <- views %>%
  summarize(
    videos = n(),
    creators = n_distinct(creator_id),
    avg_length = mean(video_length_sec),
    avg_impressions = mean(impressions_n),
    avg_watched = mean(watched_n),
    avg_watch_rate = mean(watch_rate, na.rm=TRUE),
    avg_watch_share = mean(avg_watch_share, na.rm=TRUE),
    avg_watch_seconds = mean(avg_watch_seconds_when_watched, na.rm=TRUE)
  )

print(overview)
```

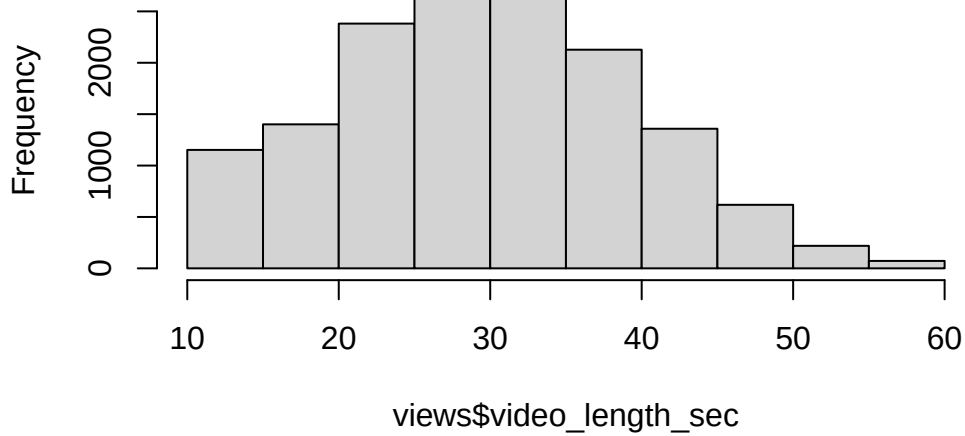
	videos	creators	avg_length	avg_impressions	avg_watched	avg_watch_rate
1	15000	200	30.08247	29.76147	20.4512	0.6946644
	avg_watch_share	avg_watch_seconds				
1	0.3794281	16.42387				

The histogram shows how video lengths are distributed. We also found `r nrow(goodvideos)` videos with a watch rate higher than 80%.

```
views$watched_pct <- views$watched_n / views$impressions_n * 100

hist(views$video_length_sec)
```

Histogram of views\$video_length_sec



```
goodvideos <- subset(views, watch_rate > 0.8)
nrow(goodvideos)
```

```
[1] 2458
```

Top Videos

This video got the most impressions in the whole dataset.

Top Video by Impressions

```
topvideo <- views[views$impressions_n == max(views$impressions_n), ]
print(topvideo)
```

	video_id	creator_id	video_length_sec	published_at	video_url
13801	13801	197	25	1754027509	
13801	https://www.tiktok.com/@elijahofficial7391/video/900000000000000013801				

	impressions_n	watched_n	watch_rate	avg_watch_seconds_when_watched
13801	230	156	0.6782609	14.29487
	total_watch_seconds	avg_watch_share	distinct_watchers_n	
13801	2230	0.3878261	134	
	first_impression_at	last_impression_at	first_watch_at	last_watch_at
13801	1754008389	1769541954	1754008389	1769541954
	watched_pct			
13801	67.82609			

To filter out noise, we look at videos with at least 60 impressions and rank them by their watch rate.

Top Videos by Watch Rate

```
top_watch <- views %>%
  select(video_id, creator_id, watch_rate, impressions_n) %>%
  filter(impressions_n >= 60) %>%
  arrange(desc(watch_rate))

head(top_watch)
```

	video_id	creator_id	watch_rate	impressions_n
1	3683	16	0.8833333	60
2	2418	125	0.8548387	62
3	7351	199	0.8536585	82
4	11903	176	0.8382353	68
5	5584	97	0.8311688	77
6	10372	174	0.8279570	93

Creator Performance

In this section, we evaluate creator productivity by looking at total impressions generated per creator and their average audience engagement.

```
creator_summary <- views %>%
  group_by(creator_id) %>%
  summarise(
    videos_n = n(),
```

```

    impressions_total = sum(impressions_n, na.rm = TRUE),
    watch_rate_avg = mean(watch_rate, na.rm = TRUE)
  ) %>%
  arrange(desc(impressions_total))

```

```
head(creator_summary)
```

```

# A tibble: 6 x 4
  creator_id videos_n impressions_total watch_rate_avg
    <int>      <int>          <int>          <dbl>
1         7        152          5791          0.700
2        96        184          5734          0.665
3        49        173          5430          0.719
4       196        143          5152          0.680
5       134        163          5105          0.696
6        75        165          4796          0.704

```

```

creator_engagement <- views %>%
  mutate(engagement = watched_n / impressions_n) %>%
  group_by(creator_id) %>%
  summarise(mean_engagement = mean(engagement, na.rm = TRUE)) %>%
  arrange(desc(mean_engagement))

```

```
head(creator_engagement)
```

```

# A tibble: 6 x 2
  creator_id mean_engagement
    <int>          <dbl>
1      117          0.749
2      143          0.743
3      154          0.741
4       55          0.738
5      114          0.736
6      184          0.732

```

Regression Analysis

This analysis investigates the relationship between `score_satiation_penalty` and `feed_rank`.

```
impressions <- read_csv("data/raw/impressions_clean.csv", show_col_types = FALSE)

model1 <- lm(
  feed_rank ~ score_satiation_penalty,
  data = impressions
)

summary(model1)
```

Call:

```
lm(formula = feed_rank ~ score_satiation_penalty, data = impressions)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.8631	-2.1181	-0.8676	1.2645	28.1766

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.61762	0.01901	137.67	<2e-16 ***
score_satiation_penalty	4.08746	0.05350	76.41	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.188 on 94788 degrees of freedom

Multiple R-squared: 0.05802, Adjusted R-squared: 0.05801

F-statistic: 5838 on 1 and 94788 DF, p-value: < 2.2e-16

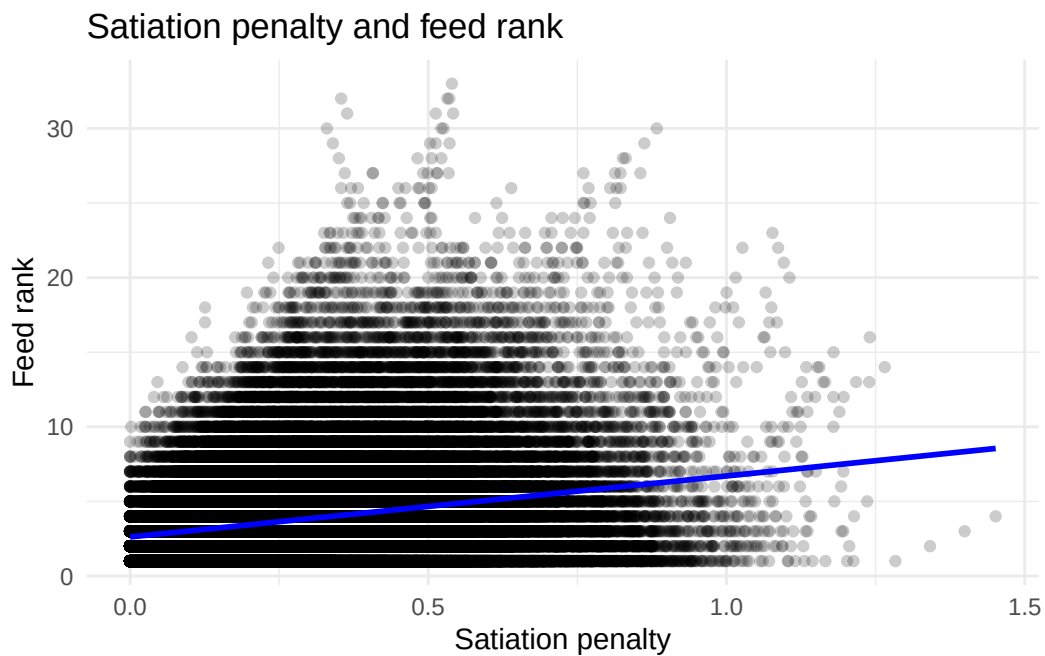
The first model shows a significant positive relationship between `score_satiation_penalty` and `feed_rank`. A one-unit increase in `score_satiation_penalty` is associated with an increase of approximately 4.09 in `feed_rank`. The model explains approximately 5.8% of the variation in `feed_rank`.

Relationship Plot

The plot above visualizes the positive relationship between `score_satiation_penalty` and `feed_rank` found in our initial regression model, complete with a linear trend line

```
ggplot(
  impressions,
  aes(x = score_satiation_penalty, y = feed_rank)
) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "blue", se = TRUE) +
  labs(
    title = "Satiation penalty and feed rank",
    x = "Satiation penalty",
    y = "Feed rank"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



The plot shows the positive relationship between `score_satiation_penalty` and `feed_rank` found in the first regression model.

Extended Regression Model

```

model2 <- lm(
  feed_rank ~ score_satiation_penalty + score_category_match + score_creator_match,
  data = impressions
)

summary(model2)

```

Call:

```
lm(formula = feed_rank ~ score_satiation_penalty + score_category_match +
    score_creator_match, data = impressions)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.8320	-2.1129	-0.8704	1.2678	28.4427

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.60612	0.01935	134.710	< 2e-16 ***
score_satiation_penalty	4.12848	0.05477	75.382	< 2e-16 ***
score_category_match	0.09961	0.01261	7.901	2.81e-15 ***
score_creator_match	-1.83569	0.21155	-8.677	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.186 on 94786 degrees of freedom

Multiple R-squared: 0.05926, Adjusted R-squared: 0.05923

F-statistic: 1990 on 3 and 94786 DF, p-value: < 2.2e-16

After adding `score_category_match` and `score_creator_match`, the relationship between `score_satiation_penalty` and `feed_rank` remains similar. The extended model explains approximately 5.9% of the variation in `feed_rank`.